

DBCPA: Dual Blockchain-Assisted Conditional Privacy-Preserving Authentication Framework and Protocol for Vehicular Ad Hoc Networks

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Abstract—Vehicular ad hoc networks (VANETs) connect all vehicles through wireless channels. They provide extensive real-time traffic information services that improve driving safety and traffic management efficiency. However, VANETs are vulnerable to security attacks because of the open wireless nature of their communication channels. Most security mechanisms for traditional VANETs are centralized and have certain limitations in satisfying security requirements, such as anti-single-point failure, distributed security authentication of messages, and privacy preservation in VANETs. To address these issues, herein, we propose a dual blockchain-assisted conditional privacy-preserving authentication framework and protocol for VANETs. The identity authentication and privacy preservation of vehicles in VANETs can be realized without relying on a centralized trusted third party. The proposed scheme also allows for the conditional tracking of illegal vehicles. The decentralized dynamic revocation of illegal vehicles can be realized through smart contracts, rendering the scheme efficient and scalable. We implement this scheme in an Ethereum test network to demonstrate its feasibility and conduct an in-depth security analysis and comprehensive performance evaluation of the proposed scheme. The results demonstrate that the proposed scheme is an effective solution for the development of a decentralized authentication system for VANETs.

Index Terms—Authentication, dual blockchain, smart contract, vehicular ad hoc networks

1 INTRODUCTION

VEHICULAR ad hoc networks (VANETs) are considered a promising method for next-generation intelligent transport systems. VANETs are distributed, self-configuring dynamic networks [1], [2]. Each vehicle in a VANET is equipped with an on-board unit (OBU) to communicate with other vehicles in the vehicle-to-vehicle (V2V) mode. V2V and vehicle-to-infrastructure (V2I) communications are realized

through a dedicated short-range communication (DSRC) protocol. Roadside units (RSUs) can also transfer communication information to the traffic control center (TCC) to further facilitate communication management and consulting services. This is similar to a set of communication patterns in the environment of traditional VANETs, as depicted in Fig. 1. Real-time information in VANETs can help vehicles or TCC take immediate action [3]. With improved perception, computing, and communication capabilities, VANETs have become a key architecture [4] for enhancing non-safety-related traffic experiences (e.g., entertainment) and safety-related facilities (e.g., traffic accidents, congestion, and sudden braking). In recent years, VANETs are receiving increasing attention from academia and industry.

1.1 Motivation

According to reports, the market value of VANETs will reach \$225,158 million in 2025 [5]. However, VANETs, as modern intelligent transportation system technology with promising prospects, also face severe data security and privacy challenges. First, VANETs data have significant economic value; as a result, data theft, attacks, and abuse have become increasingly serious issues [6]. Hence, it is a crucial requirement for VANETs to conduct a security authentication of transmitted information. Second, VANETs have the characteristics of distributed operations, limited bandwidth, node mobility, and dynamic network topology [4]. However, most security mechanisms for traditional VANETs are centralized and have certain limitations in satisfying security requirements, such as anti-single points of failure, distributed security authentication of messages, and privacy

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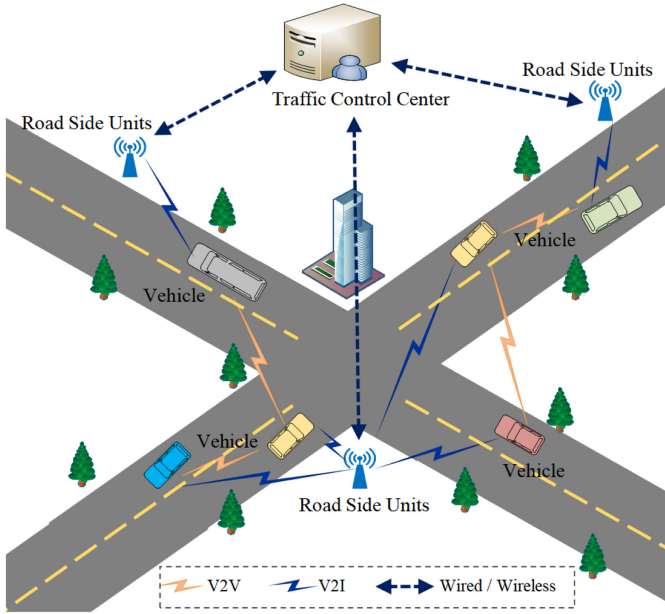


Fig. 1. A way of communication patterns in traditional VANETs.

preservation. Therefore, achieving autonomy, completely liberating trusted third parties from cumbersome authentication processes, and being resistant to single points of failure are other important requirements of VANETs. The motivation of this paper is to enhance the VANET security requirements such as VANET autonomy, anti-single point of failure, message distributed security authentication, and privacy protection, etc.

Recently, privacy-preserving authentication protocols have been proposed [7], [8], [9], [10], [11], [12], [13], [14]. Some protocols depend highly on centralized servers. Moreover, schemes based on public key infrastructure (PKI) require an authentication authority, which may experience issues with certificate management, such as certificate storage, certificate renewal, etc. Furthermore, certificate management is costly. The identity-based solution depends on the key generation center (KGC); therefore, it is easily affected by key escrows. A hybrid solution that combines the two types can alleviate these shortcomings, but the scalability in practical applications is still lacking [12].

Blockchain identifies, disseminates, and documents an intelligent peer-to-peer (P2P) network through a distributed database. The flat topology of blockchain provides network users with the benefits of autonomy, decentralization, immutability, and contractability. This is the underlying technology of Bitcoin and has been proven to have many remarkable security features [15]. Blockchain technology has been considered a potential solution for the trust and automatic inspection challenges of VANETs. Hence, blockchain attracts considerable attention as a means of enhancing the security and decentralization of VANETs [16].

Based on the problems mentioned above, this paper proposes a conditional privacy-preserving authentication scheme based on dual blockchain. This framework imparts the following advantages to VANETs. First, the dual blockchain-assisted VANET ensures that the transmitted information is reliable even without a centralized third party, thereby enhancing the security and decentralization of

VANETs. Second, it generates pseudonym identities for vehicles, updates the identities regularly, and enhances the privacy and unlinkability of the vehicles. Finally, the distributed dynamic revocation of illegal vehicles is realized using a smart contract, allowing for autonomous distributed vehicle revocation.

1.2 Contribution

We summarize the main contributions of the research content in this paper as follows.

- 1) First, a dual blockchain-assisted conditional privacy-preserving authentication framework is proposed. The proposed framework aims to enhance the decentralized features of a VANET and resolve issues related to centralized authentication and single points of failure. The proposed framework ensures that the transmitted information is reliable even without a centralized third party. In addition, privacy-preserving issues under this framework are considered.
- 2) Second, the distributed dynamic revocation of illegal vehicles is realized via a smart contract, providing the autonomous capability of distributed vehicle revocation. Expandability and robustness can be guaranteed through all nodes, rather than centralized control, which eliminates the several-for-one communication flow and single-point of failure. Therefore, it enhances the decentralized operation and improves the security of VANETs.
- 3) Finally, we conduct a security analysis, security proof, and performance analysis of the proposed scheme. The results verify the feasibility, security, efficiency, and scalability of the proposed scheme.

1.3 Organization

The remainder of this paper is organized as follows. Section 2 discusses related work on the privacy-preserving authentication schemes of VANETs. Section 3 defines the scheme model and its design security goals. Section 4 introduces the basic building blocks used in this study. Section 5 presents the details of the proposed dual blockchain-assisted privacy-preserving authentication scheme. In Section 6, we evaluate the security of the proposed DBCPA scheme. Section 7 presents an experimental evaluation of the proposed DBCPA scheme. Finally, concluding remarks are present in Section 8.

2 RELATED WORK

In recent studies, many researchers have proposed different privacy-preserving authentication schemes for VANETs. Here we roughly divide these studies into the following three categories. The first category is PKI-based. The second category is the authentication process of identity-based cryptography (IBC). The third category is solutions based on blockchain technology.

In traditional PKI-based cryptosystems, a certificate authority (CA) issues certificates to users. Reliable sources of information are ensured by digital signatures by both parties of the communication. Raya et al. [17] also proposed

a PKI-based security and privacy protocol. Wasef et al. [18] proposed in Raya et al. [17], that all vehicles need to load a large amount of anonymous public and private key pairs in advance, and the trusted authority have to also store the full anonymous certificate of each connected vehicle, which results in a certificate management is inefficient. Calandriello et al. [19] proposed a combination of baseline pseudonyms and group signatures to generate new pseudonyms, and optimized the efficiency and robustness of the hybrid scheme, simplifying the certificate management procedure. Yu et al. [20] constructed an extended pseudonym-changing region by using a pseudonym interaction strategy and an integrated group signature mechanism. Vehicles in this area are allowed to continuously exchange their pseudonyms, which better protects the location privacy of vehicles. Furthermore, Ali et al. [21] proposed a certificate-less encryption system and a PKI-based hybrid encryption scheme based on conditional privacy-preserving. Nevertheless, almost all PKI-based schemes suffer from the following three problems. First, to protect the vehicle's privacy, it has to store a considerable number of certificates and keys. Second, trusted third parties need to store all relevant certificates, which takes up a lot of storage space. Finally, the management of all the certificates would incur high computational and communication costs [12].

In the IBC scheme, a KGC issues a private key for each user based on the user's identity. The user's identity is also the user's public key. This IBC scheme can naturally associate public keys with identities, making entity identification easier [22]. Based on this, Zhang et al. [23] proposed an identity-based batch signature authentication scheme. IBC is used to generate the private key of the fake identity. This scheme does not require certificates, thus significantly reducing the transmission overhead. Chim et al. [24] propose a software-based solution that utilizes two shared secrets to satisfy privacy requirements. This enhances the privacy of the scheme of Zhang et al. [23]. But Horng et al. [8] pointed out that the SPECS method proposed by Chim et al. [24] is still vulnerable to impersonation attacks. To overcome the security flaws of previous works, Ali et al. [25] proposed an efficient identity signature scheme with conditional privacy protection. It uses common one-way hash functions and elliptic curve encryption in its scheme. The scheme provides a batch signature authentication way, where each vehicle can simultaneously verify enormous messages. Compared with the PKI-based scheme, the IBC scheme has better improvements, but they suffer from key escrow issues and the inherent drawback of requiring secure communication channels [12].

Lately, with the advent of blockchain technology, its decentralization and immutability have attracted the attention of researchers [26]. Some researchers have tried to apply blockchain technology to VANETs. Use this method to build a decentralized network security model. Raw et al. [27] propose a blockchain-based Internet of Vehicles (IoV) accident forensics framework, which stores data in entities in a distributed manner. The purpose is to provide law enforcement agencies with credible evidence about traffic accidents. Shao et al. [28] introduced a new framework of IoV combined with blockchain, which can be applied to the identity authentication scheme in IoV. It simplifies the

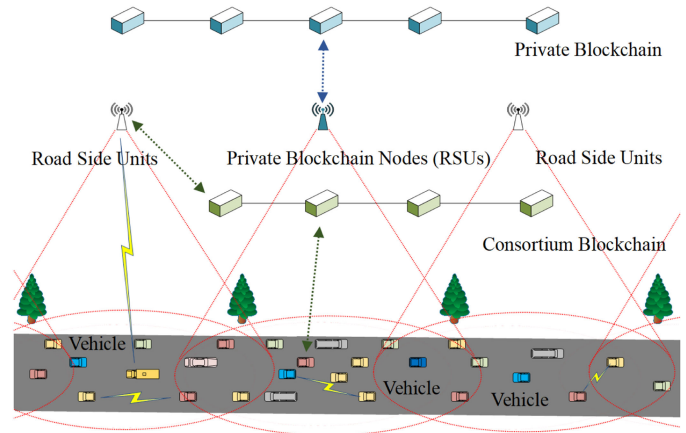


Fig. 2. Network Model.

distributed key management process in heterogeneous IoV based on blockchain technology. Reduced key transfer time over heterogeneous networks. Lu et al. [29] proposed a new BCPPA protocol that uses a fusion approach of blockchain and Merkle Patricia trees, enabling the protocol to have efficient certificate revocation and privacy protection. Zheng et al. [30] designed an identity-based BCPPA protocol using the pseudonym technique. However, Lin et al. [31] pointed out that the generating an anonymous certificate process, Lu et al. [29] requires frequent interaction between the vehicle and the certification authority, and Zheng et al. [30] faced the requirement of ideal hardware and can not withstand the leakage of the certificate authority. The VANETs scheme based on blockchain technology in the above solution lacks consideration of security and privacy. Simultaneously, the use of blockchain to realize the decentralized operation of the VANETs is still lacking, and the revocation of illegal vehicles is not automated and distributed enough.

3 PROBLEM FORMULATION

In this part, we introduce the network model of the scheme, the dual blockchain in DBCPA, and the associated security requirements.

3.1 Network Model

The network model of the framework in this paper is a three-layer model structure, as shown in Fig. 2. The upper-layer consists of a trusted authority (TA) and a private blockchain, the middle layer is the RSU (including the RSU node of consortium blockchain and private blockchain), and the under-layer is the vehicle (OBU) and consortium blockchain. We will define the three participants in this network model below.

- 1) *TA*: It is a trusted third party with considerable computing, communication, and storage capabilities, and it is generally believed that TA will not collude with other parties. TA is responsible for system initialization, entity registration, smart contract deployment, private blockchain maintenance, malicious vehicle tracking, and vehicle revocation.
- 2) *Vehicle (OBU)*: Serve the drive and all vehicles are equipped with an OBU, which contains a tamper-proof processing unit and supports the DSRC protocol. The

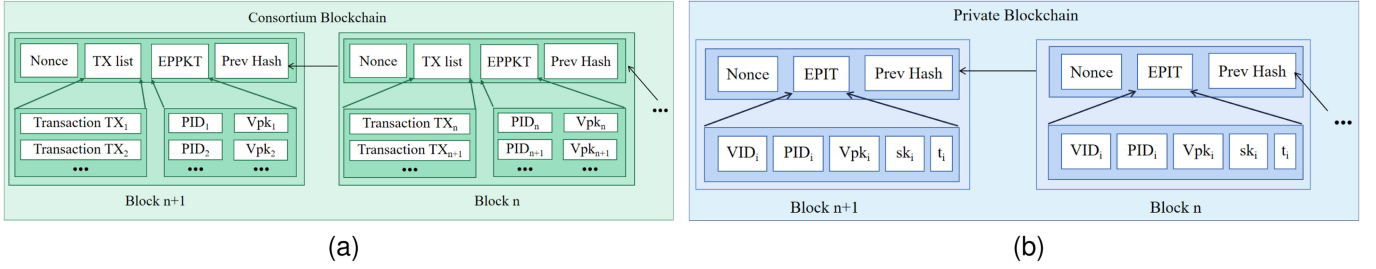


Fig. 3. Block structure of private blockchains and consortium blockchains. (a) Structure of the under-layer consortium blockchain. (b) Structure of the upper-layer private blockchain.

vehicle communicates wirelessly with other parties in the VANET through the OBU. As a validator of the consortium blockchain, the vehicle is a lightweight node that only needs to store the block header of the consortium blockchain.

- 3) *RSU*: It is the roadside infrastructure that communicates with the OBU using the DSRC protocol. Initialize pre-selected trusted RSUs as private blockchain management nodes, which can query blockchain information and periodically change pseudonyms for vehicles. Other RSUs act as miners of the consortium blockchain, aggregating communication messages between vehicles and storing all transaction data on the blockchain. As the middle layer of the network model, RSUs act as bridges to ensure that the transmitted information is trusted even without a centralized third party. And through the vehicle request, the smart contract of the private blockchain can be called to execute the distributed dynamic revocation process of the offending vehicle. The relevant information of the revoked vehicle will be deleted from the dual blockchain and broadcast on the whole network.

Remark: With the development and application of 5G technology by governments around the world, 5G, as a new technology, has the characteristics of fast speed, slow delay, wide coverage and support for communication between devices. This technology creates huge opportunities for mobile ad hoc networks, especially VANETs [32], [33], [34]. In this paper, we use 5G technology to support our proposed DBCPA approach. Because it can integrate multiple radio access technologies into cellular system architectures. Reusing available cellular network infrastructure will reduce the cost of deploying vehicle networks. The technology can also provide peak data rates of up to 20 Gb/s, with average data rates greater than 100 Mb/s, to support high data rate services [35]. The capacity of 5G has increased by a factor of 1000, enabling more efficient V2V communication [36], and reliable connectivity and low latency (about 1 ms) also provide sufficient support for the transmission of messages [37].

3.2 Dual Blockchain in DBCPA

- 1) *Blockchain*: The upper layer of the network model enables a private blockchain to increase access control and security. The TA is used as a miner to store the association relationship between the vehicle's "VID_i-PID_i-Vpk_i-sk_i-t_i". Only TAs can deploy,

insert, update and revoke smart contracts, and other licensees (trusted RSU private blockchain nodes) can access on-chain information. And TAs can verify the new state of all transactions and smart contracts. After blocks are verified by TAs, they can be uploaded to a private blockchain based on the practical byzantine fault tolerance (PBFT) [38] consensus mechanism. The lower layer of the network model uses the consortium blockchain as the decentralized underlying architecture of DBCPA. RSU acts as the miner to store all vehicle pseudonymous identities and corresponding public keys to automatically and timely feedback on the public key query of vehicles. Meanwhile, V2V and V2I transactions are aggregated, and the vehicle participates in block authentication through the consensus mechanism, safely handles transactions and smart contract status between multiple consensus nodes, and adds blocks to the blockchain. Permitted nodes can look up these states when needed to get the information the node needs in the consortium blockchain.

- 2) *Smart contract*: For consortium blockchains, it loads an entity pseudonymous public key table (EPPKT). It is used to aggregate the relationship between the current pseudonym identity of all vehicles and the corresponding public key, and to automatically and timely feedback on the public key query of the vehicle, as depicted in Fig. 3a. For private blockchains, it loads an entity private information table (EPIT), as shown in Fig. 3b. It is used as a "VID_i-PID_i-Vpk_i-sk_i-t_i" association that aggregates all vehicles and responds to the private blockchain node RSU. A new pseudonym identity and corresponding public and private keys can be obtained for the vehicle periodically, and the EPPKT can be updated. Moreover, when the revocation application of a vehicle reaches the threshold, the smart contract can automatically execute the dynamic revocation of malicious vehicles. In particular, smart contracts are on-chain code stored in the blockchain, executed by authorized entities, and participating in the authentication. Therefore, smart contracts guarantee the reliability of calculation results.

3.3 Design Goals

According to the latest research results [15][39], [40], [41], [42], the blockchain-assisted VANETs authentication scheme should meet the following security requirements.

- **Single registration:** To bring convenience to users, VANETs's authentication scheme should support single registration, where each vehicle is registered only once and then it sends information to other parties on the road.
- **Message authentication:** To ensure that the received message can be credible, the RSU and vehicle should be able to verify the validity of the message originator's identity, the timeliness, and integrity of the message, to confirm the reliability of the received message.
- **Privacy protection:** The RSUs or other vehicles should not be able to see the other vehicles' real identities, and any adversary should not be able to obtain the users' real identities even by analyzing the intercepted text.
- **Traceability:** When malicious behavior occurs, such as sending false information to mislead others, tampering with security messages, etc., the TA should have an efficient method to track the vehicle's real identity.
- **No online registration center:** To alleviate the registration center's overhead in the certification process, the scheme proposed in the study should not rely on the online registration center during the authentication stage.
- **Unlinkability:** To prevent the vehicle's driving route from being tracked or attacked by malicious attackers, it is impossible to link two pieces of information from the same vehicle.
- **Resistant to cyber attacks:** In general, VANETs authentication systems based on blockchain technology should be resistant to offline password guessing attacks, replay attacks, vehicle impersonation attacks, and distributed denial of service (DDoS) attacks.

3.4 Threat Model

In the proposed scheme, the vehicle and alliance blockchain node RSUs are considered untrusted entities, the private blockchain node RSUs are considered semi-trusted with a medium security level, and the root TA is considered fully trusted by all entities. The adversary could be a vehicle, a consortium blockchain node RSU, or a network intruder. According to the behavior of the opponent, the attacks launched by the opponent can be divided into two types, that is, passive attack and active attack. For passive attacks, an attacker continuously monitors the VANET communication channel to track the vehicle or break the confidentiality of the message. To launch an active attack, an attacker may attempt to modify or fabricate a message.

4 BUILDING BLOCKS

In this part, we introduce some basic methods and techniques used in DBCPA.

4.1 Elliptic Curve Cryptosystem

A cryptosystem based on rational point groups of elliptic curves over finite fields is called an elliptic curve cryptosystem [43]. Because of the short key and security assumptions of the elliptic curve discrete logarithm problem, it brings

the advantages of small storage space, high security, lightweight, and fast processing speed to the elliptic curve. In this study, we design an elliptic curve-based DBCPA scheme, and its related definitions are as follows.

An elliptic curve E over a finite field F_p is a finite cyclic group that satisfies $y^2 = x^3 + ax + b \pmod{p}$ and contains an infinite point $\mathcal{O}$, where $a, b \in F_p$, $4a^3 + 27b^2 \neq 0$ [44]. The points of the elliptic curve E satisfy the abelian group using the point addition "+" operation [45]. We describe its operational definition below: Suppose there are any two points on the elliptic curve, P and Q , respectively, and draw a straight line passing through P and Q . If P and Q coincide with the same point, the line is tangent to point P . The third point where this line intersects the elliptic curve E , denotes it as $-B$. This point $-B$ is symmetrical about the X-axis, represented by B , then point B must also be on this elliptic curve, that is, $E(B + (-B) = 0)$. According to the definition mentioned above, the formula for adding points on the E can be expressed as $P + Q = B$. The addition of the same k points on E , that is, $P + P + \dots + P$ is also recorded as $k \cdot P$, which is called the dot product [46].

Cryptographic systems constructed using the elliptic curve discrete logarithm problem (ECDLP) and the computational elliptic curve Diffie-Hellman problem (ECDHP) are widely used in encryption, authentication, and other cryptographic protocols [47].

4.2 Blockchain

In most common cases, blockchain is viewed as a secure, immutable, distributed ledger of data, in which data is recorded in blocks in chronological order [12]. When a blockchain allows anyone to join/leave a network and anyone within the network can maintain it, it is called a public blockchain. Such as Bitcoin and Ethereum [48]. Common consensus mechanisms for public blockchains are Proof-of-Work (PoW) [49] and Proof-of-Stack (PoS) [50]. However, when a blockchain is maintained by trusted nodes, it is known as a private blockchain or a consortium blockchain, e.g., Hyperledger [51]. Private blockchains use consensus mechanisms such as PBFT [38] and RAFT [52]. In the proposed scheme, we utilize the private blockchain as the core data storage structure of the network's upper-layer architecture. Use the consortium blockchain as the core data storage structure of the network under-layer architecture. Apply distributed ledgers with encryption technology to P2P networks to store, share, manage, track and monitor data to satisfy user privacy and security concerns.

4.3 Smart Contracts

A smart contract is a computerized transaction protocol that executes the terms of the contract, referring to "a set of digitally defined commitments, and also includes protocols that enable parties to a contract to enforce these digitally-defined commitments" [31]. Every smart contract has an address needed to trigger it. Functions defined in smart contracts can be invoked by transactions or other smart contracts. The overall objective is to meet the contractual conditions agreed upon by the parties. In general, it combines with blockchain technology to provide some attractive properties, namely self-execution, immutability, self-validation, self-execution, Authorized licensed use limited to: Institute of Information Engineering CAS. Downloaded on March 28, 2026 at 14:48:39 UTC from IEEE Xplore. Restrictions apply.

and decentralization [12]. This is the reason why smart contracts can be automatically executed and deployed in a decentralized network.

In the proposed DBCPA scheme, we leverage smart contracts to provide an application binary interface (ABI) for EPIT management services in a private blockchain. There are seven main algorithms for smart contracts deployed in DBCPA. Four of these algorithms are used to declare the relevant basic function methods in smart contracts, which are summarized below.

- 1) Data insertion algorithm: When a new vehicle enters VANET and initiates a registration request to TA to join, TA generates the security data required for communication for the vehicle, and after passing the consensus of the blockchain node, the algorithm of the smart contract is called to insert the data that needs to be updated regularly into the EPIT of the blockchain.
- 2) Data update algorithm: When the safety data stored in the blockchain is about to be exhausted, TA generates new safety data needed for vehicle communication. After passing the consensus of the blockchain nodes, TA invokes the algorithm of the smart contract to update the new security data generated for vehicle communication to the EPIT of the blockchain.
- 3) Data query algorithm: When a vehicle is about to update the security data required for related communication, the blockchain node calls the algorithm of the smart contract to query the new security data required for the communication for the vehicle, and saves it in the blockchain node for use by the vehicle.
- 4) Data deletion algorithm: When a new vehicle leaves VANET and initiates a cancellation request to TA, TA passes the consensus of the blockchain node and calls the algorithm of the smart contract to delete the relevant data of the vehicle from the EPIT of the blockchain.

The other three algorithms are used to declare the relevant non-basic function methods in smart contracts, see the algorithm 1, 2, and 3, which is introduced below.

Algorithm 1. Entity Threshold Settings

Input: address vehicleUser, uint256 txid
Output: bool result
//The threshold value changes dynamically according to the number of vehicles, flow rate and other factors.

- 1: **require** (addressID.status is **true**);
- 2: **if** msg.sender is not TA **then**
- 3: **throw**;
- 4: **end if**;
- //This is the mapping relationship between the number of the complained vehicle and the pseudonym of the complained vehicle.*
- 5: mapRespondent[vehicleUser]=txid;
- 6: **if** Entity threshold settings is successful **then**
- 7: **return true**;
- 8: **else**
- 9: **return false**;
- 10: **end if**;

- 1) Algorithm 1 states that when a new vehicle enters VANET, TA will invoke the smart contract algorithm to dynamically set the automatic vehicle revocation threshold based on the number of vehicles, traffic, and other factors.
- 2) Algorithm 2 defines the ABI for TA and RSU. When an entity complains about a malicious vehicle, the smart contract is activated to add the vehicle's complaint record after the consensus process of block nodes.
- 3) Algorithm 3 states that TA/RSU uses smart contracts to determine whether the complaint records of certain malicious vehicles reach the threshold. If the threshold is reached, the revocation algorithm for malicious vehicles is automatically executed, and the revocation-related information is returned to the private blockchain node.

We predefine the query rights of EPIT in the private blockchain appropriately. For DBCPA, we require that only private blockchain nodes (TA and part of trusted RSUs) can query EPIT. This setting can provide conditional privacy-preserving for vehicles, and improve the traceability and non-connectivity of the VANETs.

Algorithm 2. Complaints Against Malicious Vehicles

Input: address addressUser, uint256 complaintCounter
Output: uint256 txid

- 1: **require** (addressID.status is **true**);
- 2: **if** msg.sender is not TA/RSU **then**
- 3: **throw**;
- 4: **end if**;
- 5: Counter=ExistingCounter+ComplaintCounter;
//This is the mapping relationship between the complaint vehicle number and the pseudonymous identity of the complaint vehicle.
- 6: mapComplainant[addressUser]=CurrentCounter;
- 7: **return** complaint(msg.sender);

Algorithm 3. Judgment Threshold Automatically Executes Vehicle Revocation

Input: address target, uint256 CurrentCounter
Output: uint256 txid

- 1: **require** (addressID.status is **true**);
- 2: **if** Counter $\geq$ mapRespondent[target].Threshold **then**
- 3: removeEPIT(target);
- 4: **return** judgment(msg.sender);
- 5: **end if**;

4.4 Consensus Mechanism

In the proposed scheme, we adopt the PBFT [38] consensus mechanism for block consensus. As shown in Fig. 4, in the private blockchain, the TA is the miner node, and the trusted RSU is the consensus node. In the alliance blockchain, the RSU is the miner and the vehicle is the consensus node, the consensus process is as follows. First, other nodes verify the validity of the block after receiving the data block broadcast by the miner and broadcast its authentication

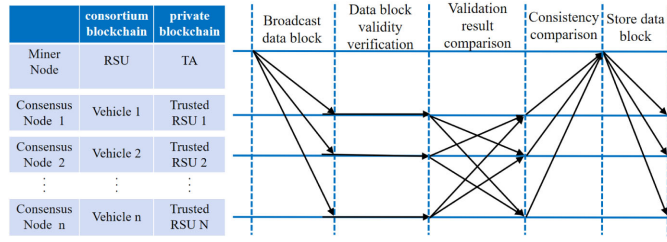


Fig. 4. Consensus process on blockchain.

results and signatures to each other in a distributed manner. Second, after each node receives the authentication result, it compares the result with other nodes and feeds the result back to the miners. Finally, the block is stored on the blockchain if 2/3 of the nodes agree to the block. In the proposed scheme, in the consensus process of the private blockchain, TA and RSU act as full nodes to store the complete information of the block. In the consensus process of the consortium blockchain, the miner RSU acts as a full node to store the complete information of the block. The vehicle acts as a lightweight node to store block header information for data query and authentication.

5 SCHEME DESIGN

In this section, we detail the information on the designed DBCPA scheme. The program makes up of seven parts: system initialization, smart contract deployment, vehicle registration, login and authentication, pseudonym identity update, vehicle revocation, and password change. Common symbols and annotations in this paper are depicted in Table 1.

5.1 System Initialization

In system initialization phase, TA generates system parameters and initializes a private blockchain and a consortium blockchain. TA selects a large prime number q_1 , a random number $x \in \mathbb{Z}_q^*$ as the key, and selects a secure hash function $H_0: \{0, 1\}^* \rightarrow \{0, 1\}^l$. TA shares $\Gamma = \{q_1, Z_q^*, H_0\}$ with the vehicle during the registration phase.

ECC Initialization: TA generates the system parameter $ECCparams = \{q, a, b, n, P, P_{pub}, H_1, H_2\}$, where the finite field $\mathbb{F}_p(q, a, b) \in \mathbb{F}_p$, defines a non-singular elliptic curve E for the equation $y^2 = x^3 + ax + b \bmod p$ satisfying $4a^3 + 27b^2 \neq 0$. P is the generator of the n -order additive group $\mathbb{G}$, which makes up of all points on E and points at infinity $\mathcal{O}$; there are system public key $P_{pub} = s \cdot P$ and system key $s \in \mathbb{Z}_n^*$; $H_1: \{0, 1\}^* \rightarrow \{0, 1\}^\lambda$, $H_2: \{0, 1\}^* \rightarrow \mathbb{Z}_n$, are two cryptographic hash functions that have fixed output ranges.

Blockchain Initialization: The upper layer of the network model enables a private blockchain to increase security, store EPIT, and manage the private blockchain by TA as a miner. The block is verified through the consensus process of the private blockchain nodes, and then the smart contract is called to deploy the transaction into the private blockchain. We presume that TA and part of trusted RSUs work together as private blockchain nodes, and give this party of trusted RSUs private blockchain nodes the authority to query the private blockchain. The under-layer of the network model enables a consortium blockchain, with RSUs as the miner. Manage EPPKT and aggregate V2V and V2I

 TABLE 1
Symbol Definition and Notes Table

Symbol	Definition
V2V	Vehicle-to-vehicle
V2I	Vehicle-to-infrastructure
OBV	On-board unit
RSU	Road side unit
TA	Trusted authority
ECC	Elliptic curve cryptosystem
E	Elliptic curve
$\mathbb{G}, q, P$	An additive elliptic curve group $\mathbb{G}$ with order q generated by P
s	System secret key
P_{pub}	System public key
x	System secret key for vehicle registration
$H_i(i = 0, 1, 2)$	Hash functions
VID_i	The real identity of the vehicle V_i
PID_i	Pseudo identity of the vehicle V_i
EPIT	Entity private information table
EPPKT	Entity pseudonym public key table
$PROC_{PID-generate}$	Pseudonymous identity generation processes
$PROC_{M-sign}$	Message signing processes
$PROC_{M-verify}$	Message verification processes

transactions, vehicles participate in block authentication, and add blocks to the consortium blockchain through a consensus mechanism.

5.2 Smart Contract Deployment

The TA compiles and deploys the smart contract's inputs into the private blockchain. The blockchain node is verified, after which the smart contract gets its unique address and can be invoked with a transaction with the appropriate permissions.

5.3 Vehicle Registration

For all vehicles and their owners to obtain the secret authentication factors they need to communicate, they need to register with the TA during the vehicle registration phase to obtain the required information. This stage is performed within a secure channel and the TA is registered offline.

First, the vehicle user selects a physical real identity VID_i , a password pw , the vehicle V_i selects a random number b_i , and calculates $B_i = H_0(pw || b_i)$. Then submit the VID_i together with B_i to the TA.

Second, TA calculates a pseudonym identity set $PID_i = VID_i \oplus H_1(sk_i \cdot P_{pub})$ for each vehicle, where $sk_i = H_2(VID_i || s || t_i)$ is the vehicle's key, t_i is the effective period of the current pseudonym, and the public key $V_{pk_i} = sk_i \cdot P$. TA selects random number $e_i \in \mathbb{Z}_q^*$, and calculates $A_i = H_0(x || e_i)$, $C_i = A_i \oplus B_i$, $D_i = (VID_i || B_i || A_i)$.

Finally, TA loads $\{C_i, D_i, e_i, H_0, q, Z_q^*, ECCparams, VID_i, PID_i, t_i\}$ into the OBU of the vehicle, with PID_1 as the first pseudonym.

In addition, TA uploads $\{VID_i, PID_i, V_{pk_i}, sk_i\}$ to the smart contract through the consensus process of private blockchain members and add it to the private blockchain. Through the consensus process of RSUs, TA uploads the $\{PID_i, V_{pk_i}\}$ of the vehicle to the consortium blockchain.

5.4 Login and Authentication

When a vehicle wants to broadcast traffic information to neighboring RSUs and vehicles, it takes the processes below.

- 1) User inputs VID_i and pw to OBU, OBU calculates $B_i^* = H_0(pw \parallel b_i)$, $A_i^* = C_i \oplus B_i^*$, $D_i^* = H_0(VID_i \parallel B_i^* \parallel A_i^*)$, and judges $D_i^* \stackrel{?}{=} D_i$. The OBU refuses the request if the authentication fails.
- 2) If this is not the case, it further generates a random number $r_i \in Z_n$ and computes $R_i = r_i \cdot P$, $\alpha_i = H_2(PID_i \parallel R_i \parallel M_i \parallel T_i)$ and $\theta_i = r_i + \alpha_i \cdot sk_i \bmod n$, where M_i denotes an immediate message sent at timestamp T_i . Finally, the OBU broadcasts the following messages $\{PID_i, M_i, R_i, T_i, \theta_i\}$ to neighboring RSUs and vehicles.
- 3) The timestamp T_i is used by the receiver to verify the message's freshness. If it works, the receiver queries the consortium blockchain for the record with the PID_i , requests the associated public key, and further searches the EPPKT identified by the PID_i to obtain the appropriate public key V_{pk_i} or err .
- 4) After the receiver gets the V_{pk_i} , the receiver computes $\alpha'_i = H_2(PID_i \parallel R_i \parallel M_i \parallel T_i)$ and will only accept the message if $\theta_i \cdot P = R_i + \alpha'_i \cdot V_{pk_i}$. Otherwise, it rejects the message. Since $V_{pk_i} = sk_i \cdot P$, $\theta_i = r_i + \alpha_i \cdot sk_i \bmod q$, we have $\theta_i \cdot P = (r_i + \alpha_i \cdot sk_i) \cdot P = R_i + \alpha_i \cdot V_{pk_i}$. As a result, the authentication process' correctness is certified. The RSU or the vehicle can detect and abort the message once it has been modified.

5.5 Pseudonym Identity Update

- 1) When the pseudonymous identity of the vehicle is about to expire, the preset private blockchain trusted node RSU will periodically obtain the new pseudonym PID'_i of the vehicle and the corresponding public and private keys $\{V'_{pk_i}, sk'_i\}$ from private blockchain, $\{PID'_i, sk'_i, V'_{pk_i}, t_i\}$ as messages M_i . RSU further encrypts and signs the message M_i , it generates a random number $r_i \in Z_n$ and computes $R_i = r_i \cdot P$, $\alpha_i = H_2(PID'_i \parallel R_i \parallel M_i \parallel T_i)$ and $\theta_i = r_i + \alpha_i \cdot sk_i \bmod n$, where M_i denotes an immediate message sent at timestamp T_i . Finally, the message $\{PID'_i, M_i, R_i, T_i, \theta_i\}$ is sent to the corresponding vehicle through the secure channel.
- 2) The timestamp T_i is used by the recipient vehicle to verify the message's freshness. If the timestamp is invalid it is discarded. If valid, the receiver will verify the correctness of the message, computing $\alpha'_i = H_2(PID'_i \parallel R_i \parallel M_i \parallel T_i)$ and only if $\theta_i \cdot P = R_i + \alpha'_i \cdot V'_{pk_i}$ to accept the message and update the OBU's vehicle pseudonym and its corresponding public keys and private keys. Otherwise reject the message.
- 3) At last, the RSU deletes outdated information in the private blockchain EPIT and updates the EPIT to facilitate conditional tracking of malicious vehicles later. And RSU updates the EPPKT of the consortium blockchain. Note that the above operations are performed offline.

5.6 Vehicle Revocation

When a vehicle departs the region or is waiting to be revoked, there are two different processes for performing a vehicle revocation. First, TA will terminate the present state documented in the smart contract. For the security of vehicle revocation, it is necessary to reach a consensus through the consensus process of blockchain nodes, and then the master node calls the smart contract to revoke the vehicle. Second, set a threshold (the threshold is dynamically changed by the current traffic flow and other related factors) for smart contract vehicle revocation. When the number of master nodes calling the smart contract to request to revoke the vehicle reaches the threshold, the smart contract executes the vehicle revocation algorithm $revokeEPIT(PID_i)$ to revoke the vehicle. In the meantime, the relevant tuples in the dual blockchain are deleted and fed back to the relevant master nodes. The master node then broadcasts the relevant information of the revoked vehicle to the whole network. More satisfy the distributed operation characteristics of VANETs and enhance the security of VANETs. We use the concept of mandatory access control to prevent malicious deletion or modification. This ensures that the state of the blockchain can only be changed by consensus TAs and the number of master node requests reaching a threshold.

5.7 Password Change

User sends VID_i and pw, pw' to OBU, OBU calculates $B_i^* = H_0(pw \parallel b_i)$, $A_i^* = C_i \oplus B_i^*$, $D_i^* = H_0(VID_i \parallel B_i^* \parallel A_i^*)$, judges $D_i^* \stackrel{?}{=} D_i$, if this authentication fails, OBU rejects the request. Otherwise, accept the new password pw' of the vehicle, OBU select b'_i , calculate $B'_i = H_0(pw' \parallel b'_i)$, $C'_i = A_i \oplus B'_i$, $D'_i = H_0(VID_i \parallel B'_i \parallel A_i)$, OBU replaces the tuple $\{C_i, D_i, e_i, H_0, q_1, Z_q^*\}$ with $\{C'_i, D'_i, e_i, H_0, q_1, Z_q^*\}$.

6 SECURITY ANALYSIS AND SECURITY PROOF

In this part, we discuss the security requirements that the proposed scheme can satisfy and provide security proof. It was demonstrated that DBCPA meets all of the security requirements mentioned in Section 3.

6.1 Security Analysis

- *Single registration*: During the vehicle registration stage in the DBCPA scheme, TA provides authentication information for all vehicles that join VANETs. Then after, all vehicles can obtain authentication information by themselves or through the RSU query of the consortium blockchain, and conduct mutual authentication without TA. As a result, DBCPA is capable of satisfying the requirements of a single registration.
- *Message authentication*: The vehicle or RSU can use the formula $\theta_i \cdot P = R_i + \alpha_i \cdot V_{pk_i}$ to check the validity of the received message, where $\alpha_i = H_2(PID_i \parallel R_i \parallel M_i \parallel T_i)$. Due to hash function collision resistance and ECDLP. Thence, any adversary with a probabilistic polynomial-time adversary will be unable to manufacture acceptable authentication text. The RSU or other vehicles will be able to detect any changes to the

message. As a result, DBCPA holds up secure message transmission authentication.

- *Privacy protection*: During the certification process, DBCPA protects the real identity VID_i of the vehicle with regularly updated pseudonymous identities. Since the blinding factor of $PID_i = VID_i \oplus H_1(sk_i \cdot s \cdot P)$ is a Diffie-Hellman tuple. And because of the ECDH problem, we came to the conclusion that no malicious entities have access to actionable information about the real identity. Therefore, DBCPA can effectively protect vehicles' privacy.
- *Traceability*: In the smart contract design, the mask value of the identification information contained in the smart contract is $PID_i = VID_i \oplus H_1(sk_i \cdot s \cdot P)$, where sk_i is the vehicle's key and s is the TA's master key. This represents that only the private blockchain nodes and related vehicles may see the real identity. Hence, when bad behavior occurs, private blockchain nodes quickly determine the real identity based on the private blockchain's data. Moreover, because of the data integrity given by the blockchain, this account capability may be trusted. As a result, DBCPA can thus give traceability.
- *Unlinkability*: The private blockchain nodes of RSU regularly update the pseudonymous identity of the vehicle. Here, no message recipient can discover the real identity of the sender or connect the two messages to a single entity. Therefore, unlinkability is satisfied.
- *No online registry*: According to the description of DBCPA, the RSU or vehicle obtains the required information through smart contracts deployed and executed in the blockchain. Use the obtained information to verify its signature and check the received message's validity. In this scheme, the TA is not involved in the whole authentication process. As a result, DBCPA can implement authentication without the use of an online registration center.
- *Resist offline password guessing attacks*: It is assumed that the message $\{PID_i, M_i, R_i, T_i, \theta_i\}$ can be observed by an adversary in the network by eavesdropping on the communication network. However, as analyzed earlier, there is no information regarding the password string in these intercepted messages. To put it another way, the adversary has no means of knowing if his or her guess for the password pw is correct. As a result, DBCPA is immune to offline password guessing attacks.
- *Resisting replay attack*: The vehicle sends $\{PID_i, M_i, R_i, T_i, \theta_i\}$ for message transmission and certification during the login and authentication stage. To ensure message freshness, timestamps must be used. As mentioned before, PID_i are pseudonymous identities, $R_i = r_i \cdot P$, $\theta_i = r_i + sk_i \cdot \alpha_i$ and $\alpha_i = H_i(PID_i \parallel R_i \parallel M_i \parallel T_i)$. By checking the validity of the received messages, both the RSU and the vehicle can quickly and accurately detect any replay behavior. This is because the pseudonym is temporary in the communication process, the time stamp T_i is used to

ensure the message's freshness, the random number r_i , and the security assumption of the hash function is used.

- *Resisting vehicle impersonation attacks*: During the login phase, because no adversary can create a valid login request in probabilistic polynomial time, the OBU verifies the judgment $D_i^* \stackrel{?}{=} D_i$ to identify the owner. Moreover, the attacker cannot impersonate other legitimate vehicles to broadcast messages in the network, because only broadcast messages signed by the key sk_i preinstalled in the OBU are valid messages. As a result, DBCPA is capable of successfully defending against vehicle impersonation assaults.
- *Resist DDoS attacks*: The DBCPA inherits the ability of blockchain to resist DDoS attacks. This means that illegal actions by any users who want to make modifications to transaction records and smart contracts can be blocked, and each illegal transaction cannot be written down on the blockchain. Additionally, the ABI in the smart contract automatically performs a lookup of the vehicle's public key in the blockchain even when some point in the network is blocked. Therefore, DBCPA can defend against DDoS attacks.

6.2 Security Proof

First, we explain the definition of the ECDLP.

Definition 1. For ECDLP, a group G is generated by the generator P , which satisfies $T = t \cdot P \in G$, and $t \in Z_q^*$. In this case, with a given $T = t \cdot P$, it is impossible to learn t .

The DBCPA security model also includes adversary A and challenger B , with B maintaining hash sets L_{h1} and L_{h2} . The game between A and B defines the security model of DBCPA in detail.

- 1) *Setup-Oracle*: B is the one who generates the system's private key and parameters. The system parameters are then sent to A by B when A invokes Setup-Oracle to query.
- 2) *h1-Oracle*: When A calls the h1-Oracle query, B provides a random number y to A , with B choosing the random number $y \in Z_q^*$. The tuple $\langle info, y \rangle$ is inserted into L_{h1} by B .
- 3) *h2-Oracle*: When A calls the h2-Oracle query, B provides a random number y to A , where the random number $y \in Z_q^*$ is selected by B . The tuple $\langle info, y \rangle$ is inserted into L_{h2} by B .
- 4) *Extract-Oracle*: When A uses the PID_i about the user's identification information in an Extract-Oracle query, B be produced a message tuple $\langle PID_i, sk_i \rangle$ and delivers it to A .
- 5) *Sign-Oracle*: When A uses the message M_i about the traffic situation in the Sign-Oracle query, B constructs a message tuple $\langle M_i, R_i, \theta_i \rangle$ and delivers it to A .

In the proposed scheme, A can execute existential forgery under an adaptive chosen message attack with a probability of ρ and a time limit of T . A could break the DBCPA if he/she could construct an effective login request message. The chance that A could violate the DBCPA is indicated by $Adv_{Auth}(A)$.

Theorem 1. If $\mathcal{A}$ can defeat the DBCPA scheme, then $\mathcal{B}$ can defeat the ECDLP in period T , which is estimated to be less than $\frac{120686\tilde{M}T}{\varrho}$, if $\varrho \geq \frac{10(N+1)(\tilde{N}+\tilde{M})}{q}$. The probability here is derived based on the Forking Lemma in literature [53], that is, $T' \leq \frac{84480\tilde{M}T}{\varrho} = \frac{84480\tilde{M}T}{7M} \cdot 2^k$ can be deduced from Theorem 10, $7Q \leq \frac{7\varrho}{10} \cdot 2^k$ can be deduced from Theorem 13, and $T' \leq \frac{84480\tilde{M}T}{7Q} \cdot 2^k = \frac{84480\tilde{M}T}{\frac{7\varrho}{10} \cdot 2^k} \cdot 2^k = \frac{84480\tilde{M}T \cdot 10 \cdot 2^k}{7\varrho \cdot 2^k} \cdot 2^k = \frac{120686\tilde{M}T}{\varrho}$ can be deduced from the combination of Theorem 10 and Theorem 13. And here, $\tilde{M}$ and $\tilde{N}$ are the number of times that $\mathcal{A}$ can interrogate the random oracle and sign the oracle, respectively.

Proof. Suppose $\mathcal{B}$ runs the proposed signature process given an ECDLP specimen $(P, sk_i \cdot P)$ of $sk_i \in Z_q^*$. Assume $\mathcal{A}$ is capable of defeating the DBCPA scheme. Challenger $\mathcal{B}$ solves ECDLP by executing the following query from adversary $\mathcal{A}$, running adversary $\mathcal{A}$ as a subroutine with non-negligible probability. $\square$

Setup: The input to the setup algorithm is a k , where k is a security parameter. $\mathcal{B}$ randomly chooses a number s and sets it as the private key, then calculates the public key P_{pub} , here $P_{pub} = s \cdot P$. Later, Challenger $\mathcal{B}$ sends $\{P, P_{pub}, q, h_1, h_2\}$ to $\mathcal{A}$.

h1 Hash Query: If $\mathcal{A}$ calls the h_1 query with $\langle PID_i, D_i \rangle$, $\mathcal{B}$ checks whether the tuple $\langle PID_i, D_i \rangle$ already exists in L_{h1} . If the tuple already exists in L_{h1} , $\mathcal{B}$ sends $\mathcal{A}$ the relevant value H_1 in the tuple. Or else, $\mathcal{B}$ will pick a random H_1 and enter a $\langle PID_i, D_i, H_1 \rangle$ tuple into L_{h1} . Following that, $\mathcal{B}$ gives $\mathcal{A}$ the value $H_1 = h_1(PID_i \parallel D_i)$ to $\mathcal{A}$.

h2 Hash Query: If $\mathcal{A}$ calls the h_2 query with the $\langle D_i, R_i, PID_i, M_i, T_i \rangle$, $\mathcal{B}$ checks whether the tuple $\langle D_i, R_i, PID_i, M_i, T_i \rangle$ already exists in L_{h2} . If the tuple already exists in L_{h2} , $\mathcal{B}$ sends $\mathcal{A}$ the relevant value H_2 in the tuple. Or else, $\mathcal{B}$ will pick a random H_2 and enter a $\langle D_i, R_i, PID_i, M_i, T_i, H_2 \rangle$ tuple into L_{h2} . Following that, $\mathcal{B}$ gives $\mathcal{A}$ the value $H_2 = h_2(D_i \parallel R_i \parallel PID_i \parallel M_i \parallel T_i)$ to $\mathcal{A}$.

Extract Query: If $\mathcal{A}$ calls an extract query on the identity information PID_i of user, $\mathcal{B}$ computes and examines whether the tuple $\langle PID_i, D_i \rangle$ is already stored in L_{h1} . If the corresponding pair $\langle PID_i, D_i, H_1 = h_1(PID_i \parallel D_i) \rangle$ cannot be found according to $\langle PID_i, D_i \rangle$, $\mathcal{B}$ direction $\mathcal{A}$ delivers a failure message and rejects the query. Or else, $\mathcal{B}$ evaluates sk_i and returns $\langle PID_i, sk_i \rangle$ to $\mathcal{A}$. Note that by making extract query, $\mathcal{A}$ cannot get the sk_j of the target user with PID_j .

Sign Query: If $\mathcal{A}$ makes this query on message M_i with pseudonym identity PID_i , $\mathcal{B}$ first examines whether the tuple $\langle PID_i, D_i, H_1 \rangle$ from L_{h1} . H_1 is obtained by $\mathcal{B}$ from the tuple $\langle PID_i, D_i, H_1 \rangle$. $\mathcal{B}$ then chooses two numbers at random, r_i and H_2 . Besides, $\mathcal{B}$ chooses two numbers at random, μ_i and v_i and tries again. Or else, $\mathcal{B}$ calculates $R_i = H_2^{-1} \mu_i P - \tilde{M}$ and $\theta_i = \mu_i$ and transmits $\langle M_i, R_i, \theta_i \rangle$ to $\mathcal{A}$, where $h_2(D_i \parallel R_i \parallel PID_i \parallel M_i \parallel T_i) = H_2$.

Analysis: Using the Forking Lemma [53], once $\mathcal{A}$ can produce two valid signatures $(R_i, \theta_i = H_2 \times r_i + sk_i \bmod q)$ and $(R'_i, \theta'_i = H'_2 \times r_i + sk_i \bmod q)$, and $H_2 \neq H'_2$, Then $\mathcal{B}$ can calculate sk_i from these two valid signatures.

As a result, $\mathcal{B}$ is capable of solving the ECDLP in a time that is less than $\frac{120686\tilde{M}T}{\varrho}$, where $\varrho \geq \frac{10(N+1)(\tilde{N}+\tilde{M})}{q}$. Because

this conclusion contradicts the Definition 1, the proposed

TABLE 2
Several Cryptographic Operations' Execution Times

Operation	Definition	Operation Time Symbol	Average Execution Time (ms)
<i>pa-ecc</i>	Point addition operation related to ECC	T_{pa-ecc}	0.00256
<i>sm-ecc</i>	Scalar multiplication operation related to ECC	T_{sm-ecc}	0.37374
<i>im-ecc</i>	Integer multiplication operation related to ECC	T_{im-ecc}	0.04102
<i>h</i>	Hash operation	T_h	0.0009

DBCPA scheme is secure under the adaptive chosen message attack in the random oracle model.

7 PERFORMANCE ANALYSIS

In this part, we present the performance analysis and experimental feasibility analysis of the proposed DBCPA scheme. We compared several related protocols with the protocol we studied [1][13][14]. Among them, the scheme of He et al. [1] is the original scheme of ECC-based conditional privacy-preserving authentication scheme in VANETs. The schemes of Ali et al. [13] and Sutrala et al. [14] are an improved ECC-based conditional privacy-preserving authenticated signature scheme.

7.1 Computational Overhead Analysis

We analyze the computational overhead of various methods through three procedures in this section. Namely pseudonymous identity generation processes, message signing processes including signature and key generation processes, and message verification processes. These three processes are represented by $PROC_{PID-generate}$, $PROC_{M-sign}$, and $PROC_{M-verify}$ respectively. To analyze the computational overhead of cryptographic operations, we use a C/C++ cryptographic library named MIRACL [54] to measure all involved cryptographic operations for the proposed and compared schemes. We conducted performance evaluations on a laptop running the Ubuntu 18.04.6 operating system with an Intel(R) Core(TM) i5-4590 CPU(3.30 GHz) and 4GB RAM. The results obtained are described in Table 2. We can roughly calculate the execution time associated with cryptographic operations during a successful certification process. It's worth noting that TA is offline throughout the authentication process, and we calculate the computational overhead on the vehicle side.

For the scheme of He et al. [1], the $PROC_{PID-generate}$ requires a vehicle to execute two *sm-ecc* and one *h*. Therefore, the computational overhead of the $PROC_{PID-generate}$ is $2T_{sm-ecc} + T_h = 0.74838$ ms. For the $PROC_{M-sign}$, the vehicle needs to execute one *sm-ecc* and two *h*. So the computational cost of the $PROC_{M-sign}$ is $T_{sm-ecc} + 2T_h = 0.37554$ ms. For the $PROC_{M-verify}$, the vehicle needs to execute three *sm-ecc*, two *pa-ecc*, and two *h*. Therefore, the computational overhead of the $PROC_{M-verify}$ is $3T_{sm-ecc} + 2T_{pa-ecc} + 2T_h = 1.12814$ ms. So

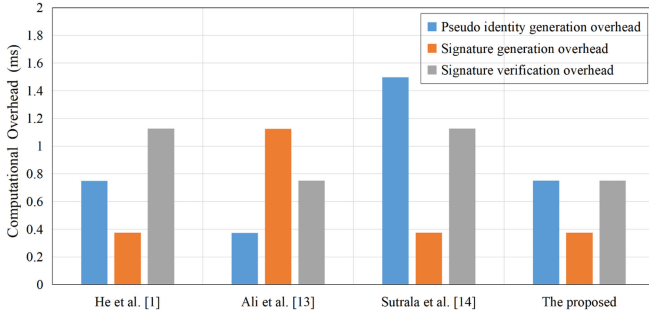


Fig. 5. Compares the computational overhead in $PROC_{PID-generate}$, $PROC_{M-sign}$, and $PROC_{M-verify}$, respectively.

the scheme of He et al. [1] has a total computational cost of 2.25206 ms for the three processes.

For the scheme of Ali et al. [13], the $PROC_{PID-generate}$ requires a vehicle to execute an $sm-ecc$. Therefore, the computational overhead of the $PROC_{PID-generate}$ is 0.37374 ms. For the $PROC_{M-sign}$, the vehicle needs to execute three $sm-ecc$, one $pa-ecc$, and one h . So the computational cost of the $PROC_{M-sign}$ is $3T_{sm-ecc} + T_{pa-ecc} + T_h = 1.12468$ ms. For the $PROC_{M-verify}$, the vehicle needs to execute two $sm-ecc$, one $pa-ecc$, and one h . Therefore, the computational cost of the $PROC_{M-verify}$ is $2T_{sm-ecc} + T_{pa-ecc} + T_h = 0.75094$ ms. So the scheme of Ali et al. [13] has a total computational cost of 2.24936 ms for the three processes.

For the scheme of Sutrala et al. [14], the $PROC_{PID-generate}$ of the vehicle needs to execute four $sm-ecc$ and two h . Therefore, the computational overhead of the $PROC_{PID-generate}$ of this scheme is $4T_{sm-ecc} + 2T_h = 1.49676$ ms. For the $PROC_{M-sign}$, the vehicle needs to execute one $sm-ecc$ and one h . So the computational cost of the $PROC_{M-sign}$ is $T_{sm-ecc} + T_h = 0.37464$ ms. For the $PROC_{M-verify}$, the vehicle needs to execute three $sm-ecc$, two $pa-ecc$, and two h . Therefore, the computational cost of the $PROC_{M-verify}$ is $3T_{sm-ecc} + 2T_{pa-ecc} + 2T_h = 1.12814$ ms. So the scheme of Sutrala et al. [14] have a total computational cost of 2.99954 ms for the three processes.

For our proposed scheme, the first $PROC_{PID-generate}$ is preloaded into OBU through TA calculation and loaded into the private blockchain, and the computational overhead on the vehicle side is 0 ms. The subsequent $PROC_{PID-generate}$ is sent to the vehicle by the RSU node of the private blockchain through a message signature, and the vehicle updates the pseudonym identity after the verification process. For the $PROC_{M-verify}$, the vehicle needs to execute an $pa-ecc$, two $sm-ecc$, and a h . Therefore, the computational cost of the $PROC_{M-verify}$ is $2T_{sm-ecc} + T_{pa-ecc} + T_h = 0.75094$ ms. Therefore, the computational overhead of the $PROC_{PID-generate}$ is $2T_{sm-ecc} + T_{pa-ecc} + T_h = 0.75094$ ms. For the $PROC_{M-sign}$, the vehicle needs to execute one $sm-ecc$ and a h . So the computational cost of the $PROC_{M-sign}$ is $T_{sm-ecc} + T_h = 0.37464$ ms. So the total computational overhead of the three processes of our proposed scheme is 1.87652 ms.

Fig. 5 shows a comparison of different approaches in terms of computational overhead. The computational overhead of our proposal is higher than that of the scheme of Ali et al. [13] in the $PROC_{PID-generate}$, but lower than that of the scheme of Sutrala et al. [14], and has almost the same computation as the scheme of He et al. [1] overhead. The proposed DBCPA scheme has the same time overhead as the

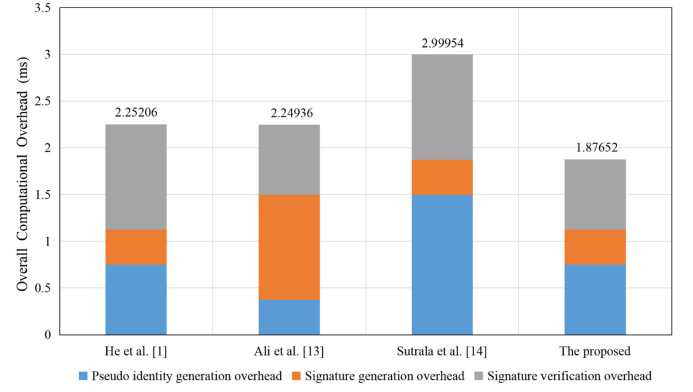


Fig. 6. The comparison of total computational overhead.

scheme of Ali et al. [13] and Sutrala et al. [14] in the $PROC_{M-sign}$ and has a smaller computational overhead than the scheme of He et al. [1]. During $PROC_{M-verify}$, our scheme has the same time overhead as the scheme of Ali et al. [13] and has the lowest computational overhead among all schemes. To sum up the above, the total computational overhead of our proposed proposal is the lowest among all schemes, as depicted in Fig. 6.

7.2 Communication or Storage Overhead Analysis

To analyze the communication overhead of cryptographic operations, we have the following settings. To achieve the 80-bit security level, we use an additive group $\mathbb{G}$, which is generated by a point P of order q on the nonsingular elliptic curve $y^2 = x^3 + ax + b \mod p$, where p and q are two prime numbers, and $(q, a, b) \in \mathbb{F}_p$. It's worth noting that when using the above settings, the traffic emergency notifications' size is restricted to 160 bits. If the size of the traffic emergency message is required to be greater than 160 bits, an elliptic curve with a larger order can be selected. For instance, curve P-256 can be selected, which supports 256-bit messages, or P-384, which supports 384-bit messages. To facilitate comparison with related schemes, we suppose that the timestamp's size is 32 bits and that the traffic-related messages are encoded as 160-bit strings (as previously stated, if we need messages of a larger size, then we can choose elliptic curves of larger order).

In the scheme of He et al. [1], the vehicle needs to broadcast $\{PID_i = (PID_i^1, PID_i^2), m_i, T_i, \sigma_i, R_i\}$, where $PID_i^1, R_i \in \mathbb{G}$, $PID_i^2, \sigma_i \in \mathbb{Z}_q^*$. As a result, He et al.'s design [1] has a communication overhead of $(320 + 160 + 160 + 32 + 160 + 32)/8 = 144$ bytes. In the scheme of Ali et al. [13], the vehicle must broadcast $\{m_i, AID_i, pk_i, \Theta_i, t_i\}$, where $AID_i, pk_i \in \mathbb{G}$, $m_i, \Theta_i \in \mathbb{Z}_q^*$. As a result, Ali et al.'s design [13] has a communication overhead of $(160 + 320 + 320 + 160 + 32)/8 = 124$ bytes. In the scheme of Sutrala et al. [14], the vehicle must broadcast $\{PID_i = (PID_i^1, PID_i^2, T_i), m_i, \delta_i = (f_i, g_i), B_i, K_i, R_i, T_i\}$, where $PID_i^1, B_i, K_i, R_i \in \mathbb{G}$, $PID_i^2, f_i, g_i \in \mathbb{Z}_q^*$. As a result, Sutrala et al.'s design [14] has a communication overhead of $(320 + 160 + 32 + 160 + 160 + 160 + 320 + 320 + 320)/8 = 244$ bytes. In the proposed DBCPA scheme, the vehicle must broadcast $\{PID_i, M_i, R_i, T_i, \theta_i\}$, where $PID_i, R_i \in \mathbb{G}$, $M_i, \theta_i \in \mathbb{Z}_q^*$. As a result, our proposed DBCPA scheme has a communication overhead of $(320 + 160 + 32 + 320 +$

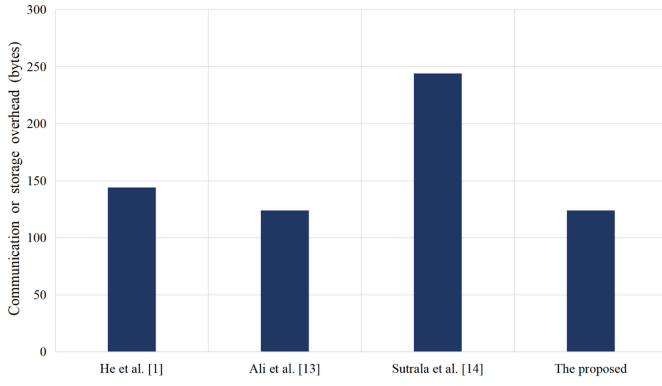


Fig. 7. The comparisons of communication overhead.

160)/8 = 124 bytes. The communication overhead comparison of these schemes is shown in Fig. 7.

We can observe that the communication and storage overhead of the proposed DBCPA scheme is the smallest. In a nutshell, the communication or storage overhead of the proposed DBCPA scheme is 86.1% of the scheme of He et al. [1] and is 51% of the scheme of Sutrala et al. [14]. There are two reasons for this phenomenon. First, we use elliptic curves in combination with blockchain. Second, the proposed DBCPA scheme minimizes the transmission cost of fake identities and signatures without sacrificing security strength.

7.3 Blockchain Practical Feasibility Analysis

To verify the feasibility of DBCPA, we execute the scheme on the Ethereum test network Rinkeby¹. Herein, Rinkeby can provide customers with free funding requests, as well as a user-friendly online interface for browsing blocks. Furthermore, we use a MetaMask-Chrome² plug-in for Google Chrome to connect Remix³ and Rinkeby in Chrome, and use them to deploy and invoke our designed smart contracts. The following are the specifics of this implementation.

First, we use MetaMask to create two accounts (TA and RSU) for our testing, the addresses are `0x93Aec-38B89aaF03A2218637ec696ba41FdD49f4C`, `0x07304fC-ea7fE206084B5CfBBb595f6f9742f1d22`. Then switch to the TA account and ask Rinkeby for 3 Ethers. In this way, the TA can initially load the relevant private information of the offline registration of the vehicle into the private blockchain. And can perform related operations such as adding, deleting, updating, and searching for vehicle private information. To ensure that the transmitted information is reliable even without a centralized third party on the VANETs. Fig. 8 depicts the results.

Second, We executed Remix as TA to deploy the smart contract to Rinkeby and get its address (i.e., `0x623a2f02-16e3BFcBf148aa2CD89E647F8AE8fB95`). We also call the vehicle threshold setting algorithm via Remix to set the threshold for revoking offending vehicles, as shown in Fig. 9.

1. <https://www.rinkeby.io>
 2. chrome-extension://nkbihfbeamgaohlefnkodbefgpgknn/home.html
 3. <http://remix.ethereum.org>

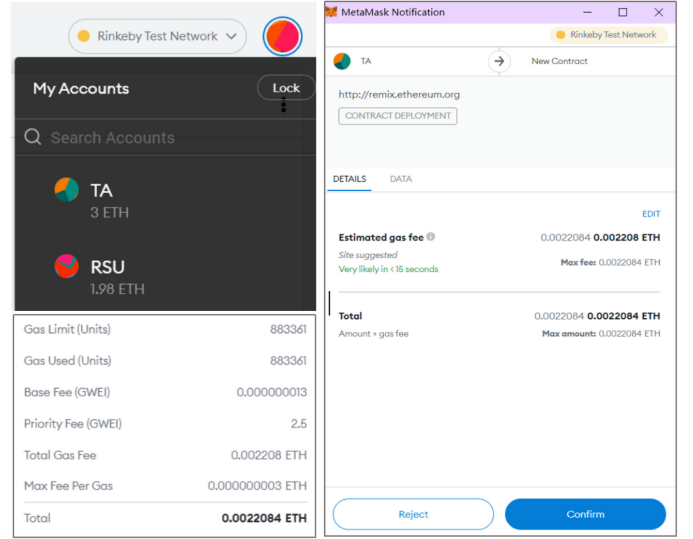


Fig. 8. TA deploys smart contracts.

Finally, when the vehicle reports the malicious behavior of the malicious vehicle to the RSU, it goes through the private blockchain nodes consensus. We simulated the private blockchain RSU nodes to add the relevant information of the reported vehicle in Rinkeby, such as the whistleblower's pseudonym, the reported person, the number of reports, etc. We switch to the RSU node account, invoke the complaint algorithm, and update the consensus reported vehicle information to the private blockchain, as depicted in Fig. 10.

Meanwhile, the smart contract calls the threshold judgment algorithm. If a reported vehicle is reported a threshold number of times, the current state recorded in the smart contract will be terminated. And the vehicle is revoked through the ABI of `removeEPIT(PIDi)`, thereby deleting the related tuples in the double blockchain, and broadcasting the vehicle revocation message to other nodes.

In addition, to test the cost in terms of transaction fees, we evaluated the gas cost of the relevant operations. And the results are shown in Table 3. The price on January 22, 2022, is 1 ETHER=2412.5 USD, and the average GAS PRICE=2.5 GWEI. Smart contracts only need to be deployed once by the TA. Table 3 shows the outcomes, the largest cost is the deployment of the smart contract, which costs about \$6.47583, but is only executed once. And the cost of other operations is less than \$1.79, which is an acceptable cost even if these operations are repeatedly called frequently.



Fig. 9. Execute the vehicle threshold setting algorithm.

Transaction Hash:
0x28a9386f3bc3b7581ca060c0203f648510202b67af42a9e225a208832973a3

Status:
Success

Block:
10016736
22 Block Confirmations

Timestamp:
6 mins ago (Jan-19-2022 06:06:55 AM -UTC)

From:
0x07304fcea7fe206084bd5cftbb659589742f1d22

To:
Contract 0x8cca5d62f8e248bd32a829603bf88ae93d67ebba

Value:
0 Ether (\$0.00)

Transaction Fee:
0.0000597565000262968 Ether (\$0.00)

Gas Price:
0.000000002500000011 Ether (2,500000011 Gwei)

Gas Limit & Usage by Txn:
23,906 / 23,906 (100%)

Gas Fees:
Base: 0.0000000011 Gwei | Max: 2,5000000018 Gwei | Max Priority: 2.5 Gwei

Burnt & Txn Savings Fees:
Burnt: 0.000000000000002262968 Ether (\$0.00)
Txn Savings: 0.0000000000000167342 Ether (\$0.00)

Others:
Txn Type: 2 (EIP-1559) | Nonce: 1 | Position: 6

Fig. 10. Implement malicious vehicle complaint algorithm.

TABLE 3
TA Smart Contract Gas Cost(Gas Price = 2.5 GWEI,
1 Ether = 2412.5 USD)

Operation	Gas Used	Actual Cost (ether)	USD
Deploy	1342141	0.002684282	6.47583
insertEPIT	3692975	0.000738595	1.78186
removeEPIT	304505	0.000060901	0.14692
updateEPIT	846055	0.000169211	0.40822
ThresholdSet	29911	0.000059822	0.14432
complaint	1724455	0.000344891	0.83205
judgeThreshold	334035	0.000066807	0.16117

8 CONCLUSION

In this study, we develop a dual blockchain-assisted conditional privacy-preserving authentication framework and protocol for VANETs. The proposed scheme fuses blockchain technology with cryptographic primitives. VANETs assisted by the dual blockchain can realize efficient privacy-preserving authentication even if the trusted third party is offline. The accuracy and reliability of the information transmitted in VANETs can be ensured while protecting vehicle privacy. The distributed dynamic revocation of illegal vehicles is realized via smart contracts. Expandability and robustness can be guaranteed through all nodes, rather than centralized control, which eliminates the several-for-one communication flow and single-point failure. Detailed security analysis indicates that the scheme satisfies the security and privacy requirements of VANETs. Performance analysis reveals that this scheme has certain advantages compared with related schemes. In future work, we will also explore flexible privacy-preserving mechanisms to construct a more practical scheme that solves the different security requirements of VANETs.

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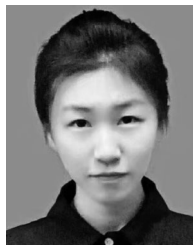
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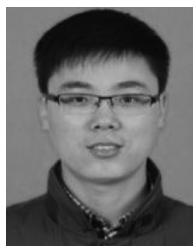
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